

Two Facets of Emotional Awareness and Psychological Distress: A Meta-Analysis

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Emotional awareness, consisting of two facets—attention to emotion and emotional clarity—plays a crucial role in psychological distress. This meta-analysis examined the distinct relationships between the two facets of emotional awareness and distress across 307 studies involving 100,612 participants. Results showed that attention to emotion was not significantly associated with distress; however, the association became significant and positive after controlling for emotional clarity. Emotional clarity showed a moderate negative correlation with distress, which remained consistent even after accounting for attention to emotion. These findings suggest that emotional clarity may be essential in reducing psychological distress, while excessive attention to emotion, when not accompanied by emotional clarity, may exacerbate psychological distress. These findings underscore the importance of fostering emotional clarity in interventions aimed at reducing psychological distress by enhancing emotional awareness.

Keywords: emotional intelligence, attention to emotions, emotional clarity, clarity of emotions, alexithymia

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Individuals utilize not only external information acquired through sensory receptors but also internally generated information (Duval & Wicklund, 1972). Being aware of one's internal states is generally helpful for the person, as it can promote self-regulation processes by provoking self-evaluation, wherein they evaluate discrepancies between their current and desired states (Carver & Scheier, 1981, 1982; Duval & Wicklund, 1972). Emotions are salient internal information that communicates to a person and those around them regarding their environments, aiding in their responses to situational demands (Keltner & Gross, 1999; Schwarz, 1990; Schwarz & Clore, 1983). Thereby, emotional awareness, the ability to identify and understand one's own emotions (Gratz & Roemer, 2004; Lane & Schwartz, 1987), is generally considered adaptive. For example, mood and emotion regulation theories (e.g., the process model of emotion regulation; Gross, 2015; the control theory of mood regulation; Larsen, 2000) posit that emotional awareness is an essential step for effective regulation.

Indeed, several meta-analyses have concluded that greater emotional awareness is generally associated with decreased psychological distress (e.g., Derks et al., 2017; Frewen et al., 2008; S. Li et al., 2015;

Sendzik et al., 2017; Visted et al., 2018). However, previous research using factor analysis (e.g., E. Coffey et al., 2003; Gohm & Clore, 2000, 2002; Palmieri et al., 2009) revealed that emotional awareness is a multidimensional construct comprising two facets: attention to emotion and emotional clarity. Attention to emotion refers to the degree to which individuals focus on and value their emotions as a source of information (Boden & Thompson, 2017), whereas emotional clarity indicates the extent to which individuals can clearly identify and label their feelings (E. Coffey et al., 2003; Gohm & Clore, 2000). Although the two have a moderate positive correlation with each other (see the meta-analysis by Boden & Thompson, 2017), they are considered relatively independent dimensions (E. Coffey et al., 2003; Gohm & Clore, 2002). Further, prior research showed that the two might exhibit distinct patterns in relation to distress-related processes. For example, individuals with heightened worry reported similar levels of emotional clarity but greater attention to emotion compared to those with low worry levels (Berenbaum et al., 2012). Therefore, it is important to investigate how the two facets of emotional awareness are associated with psychological distress, respectively, across studies.

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Among previous meta-analyses on the relationships between emotional awareness and psychological distress (e.g., Derks et al., 2017; Frewen et al., 2008; S. Li et al., 2015; Sendzik et al., 2017; Visted et al., 2018), two studies have examined the two facets separately (S. Li et al., 2015; Visted et al., 2018). S. Li et al.'s (2015) meta-analysis on alexithymia focused on a particular measure of emotional awareness (the Toronto Alexithymia Scale [TAS]) and its relation to depression.¹ The authors found that, across studies, depression had a significant moderate positive association with a lack of emotional clarity and had a small positive association with a lack of attention to emotion. In addition, Visted et al. (2018) examined differences in attention to emotion and emotional clarity between individuals with and without major depressive disorder as part of their larger research questions. Their meta-analysis, which included only two previous studies using the Difficulties in Emotion Regulation Scale (DERS), showed that individuals with major depressive disorder exhibited both lower attention to emotion and emotional clarity compared to nondepressed individuals.

Along with the findings from the two meta-analyses (S. Li et al., 2015; Visted et al., 2018), previous systematic reviews have indicated that emotional clarity is generally related to decreased psychological distress (e.g., Llamas-Díaz et al., 2022; Resurrección et al., 2014). In contrast, conclusions regarding the relationship between attention to emotion and distress remain inconclusive. Some researchers conducting meta-analyses and/or systematic reviews on this topic have shown a negative relationship between attention to emotion and psychological distress, similar to that of emotional clarity (S. Li et al., 2015; Visted et al., 2018). However, others have argued that non-significant relationships with affective well-being, including negative affect, are more commonly observed across studies (Llamas-Díaz et al., 2022) or indicated that paying greater attention to emotion may even be detrimental to psychological adjustment (Resurrección et al., 2014).

The Present Study

To the best of our knowledge, no meta-analytic study has yet separately examined the two facets of emotional awareness in their relationships with psychological distress while including all relevant studies and measures on this topic. Therefore, to address this gap in the literature and clarify the distinct relationships between these facets and psychological distress, we aimed to systematically review the literature and quantitatively summarize the pooled relationships between the two facets and psychological distress.

Based on the general consistencies in the literature as well as mood and emotion regulation theories (Gross, 2015; Larsen, 2000), we first hypothesized that emotional clarity would be negatively correlated with psychological distress. Second, we took an exploratory approach to predict the relationship between attention to emotion and psychological distress. Since emotional clarity depends on paying attention to emotion, the relationship between attention and psychological distress could resemble that of emotional clarity. Conversely, attention to emotion might be positively associated with psychological distress. Excessive attention or hypermonitoring might lead to inflexible self-regulation focused solely on affect, thereby impairing the ability to respond to contextual demands (e.g., De la Barrera et al., 2023; Swinkels & Giuliano, 1995; Tamir, 2021; Wismeijer et al., 2009). In addition, greater attention to emotion could imply hypervigilance toward one's internal experiences (e.g., Swinkels & Giuliano, 1995),

potentially resulting in a positive relationship between attention to emotion and distress.

Last, we investigated the independent role of each facet in psychological distress using partial correlation analysis. This approach addresses whether attention to emotion and emotional clarity are independently associated with distress when controlling for one another. Previous studies have shown that individuals exhibit varying patterns in these facets (e.g., Gohm, 2003), and the relationship between each facet and distress may depend on the other's level (e.g., Berenbaum et al., 2012; Sperry & Eckland, 2021). For instance, Gohm (2003) identified four distinct clusters based on emotional clarity and attention to emotion, as well as intensity, across three samples; notably, two groups differed primarily in emotional clarity despite similar attention levels—a profile replicated by others (e.g., Berenbaum et al., 2012; Sperry & Eckland, 2021). Importantly, the group with the lowest emotional clarity (and average attention) reported the greatest distress (e.g., heightened negative affect; Sperry & Eckland, 2021), suggesting that emotional clarity may be linked to distress independent of attention to emotion. Based on these findings, we predicted that the relationship between emotional clarity and distress would remain consistent even after accounting for attention to emotion, while exploring whether the relationship between attention to emotion and distress would change when accounting for emotional clarity.

Definition of Psychological Distress

Among various facets of psychological distress and related disorders examined in relation to emotional awareness, this study focuses specifically on emotional distress, as our hypotheses are grounded in theories of mood, emotion regulation, and self-regulation (Carver & Scheier, 1982; Gross, 2015; Larsen, 2000). We define psychological distress as broad subjective distress and negative emotional states, encompassing depression, anxiety, and related symptoms (including those from various anxiety disorders, posttraumatic stress disorder [PTSD], and obsessive-compulsive disorder [OCD]), perceived stress, negative affect, and other distress-related symptoms (i.e., suicidality and symptoms of borderline personality disorder [BPD]).² Most of these facets align with symptoms in the distress and fear subfactors of the hierarchical taxonomy of psychopathology model, where they share negative emotional states and distress as common features (Watson et al., 2022). Furthermore, this definition is also consistent with those used in prior meta-analyses on psychological distress, which have primarily defined it as common mental health-related difficulties and negative emotionality (e.g., Ho et al., 2022; Marsh et al., 2018).

¹ Alexithymia refers to difficulties identifying and describing one's feelings, characterized by a lack of introspection (Bagby et al., 1994), which reflects difficulties in emotional awareness (e.g., E. Coffey et al., 2003; Palmieri et al., 2009).

² Although other types of distress and related constructs have been examined in relation to emotional awareness, we included only those aligning with our definition of psychological distress. For example, we excluded well-being measures (e.g., life satisfaction) and constructs associated with distress (e.g., rumination).

Potential Moderators

We first examined the type of measurements (e.g., DERS, TAS) as a potential moderator. Prior research has suggested that the type of measurements may moderate the associations between emotional awareness and psychological distress (Boden & Thompson, 2017; Park et al., 2022). Further, different measures assess distinct aspects of emotional awareness. For instance, the TAS and the DERS focus on a person's devaluation of emotions and deficits in emotional introspection, while other measures, such as the Trait Meta-Mood Scale (TMMS) and the Mood Awareness Scale (MAS), primarily assess the extent to which a person attends to their emotions (Boden & Thompson, 2017; Salovey et al., 1995; Swinkels & Giuliano, 1995; see Table 1). Second, age and gender were tested as potential moderators, given that gender and developmental differences in emotional awareness have been reported (Mankus et al., 2016; Thayer et al., 2003). Last, we have explored study design (cross-sectional vs. longitudinal), sample characteristics (nonclinical college/community samples, psychiatric samples, vs. samples with physical conditions), and domains of psychological distress (depression, anxiety, stress, negative affect, anxiety-related symptoms vs. suicidality and BPD symptoms) as potential moderators of the relationships between emotional awareness and distress.

Method

Search Process

Articles published up to July 2025 were retrieved from APA PsycInfo, PubMed, and ProQuest Dissertations and Theses, supplemented with additional searches using Google Scholar. We searched for articles using keywords that included variants of relevant constructs (e.g., attention to emotion, emotional clarity), specific subscales for each construct (e.g., externally oriented thinking) combined with full scales (e.g., TAS), and domains of psychological distress (e.g., depression, anxiety). Specific subscales were searched in combination with full-scale names, because including only the full-scale names yielded too many irrelevant articles that did not assess each facet of emotional awareness separately. The detailed keywords used for search are presented in Supplemental Material A.

Duplicate records from multiple databases were removed. When two studies utilized the same data set, the earlier published study was selected. The published version was chosen if an unpublished dissertation was later published. For articles examining constructs of interest but not reporting effect sizes, we contacted the first and/or corresponding authors to obtain the information. Due to the large number of such cases, we limited our outreach to authors with at least two relevant articles. Among 22 contacts, one author provided the requested data, which was incorporated into our analyses. The flowchart of our search process is presented in Figure 1.

Eligibility Criteria

We included articles that (a) reported correlations between attention to emotion and psychological distress and/or between emotional clarity and psychological distress and (b) were written in English. Based on a previous meta-analysis of these two facets (Boden & Thompson, 2017), the following measures were included for each facet. For attention to emotion, we included the attention subscale of the TMMS, the monitoring subscale of the MAS, the lack of

awareness subscale of the DERS, and the externally oriented thinking subscale of the TAS, and their variants. In addition, the voluntary attention subscale of the Attention to Emotion Scale (Huang et al., 2013),³ which was developed based on the TMMS and the TAS, and the attention to feelings subscale of Skalen zur Wahrnehmung eigener und fremder Gefühle (WEFG; Lischetzke et al., 2001) were included.

For emotional clarity, the clarity subscale of the TMMS, the Emotional Clarity Questionnaire (a modified measure of the TMMS-clarity subscale for youth; Flynn & Rudolph, 2010), the labeling subscale of the MAS, the difficulty identifying/describing subscales of the TAS, the lack of clarity subscale of DERS, the clarity of feelings subscale of WEFG, and their variants were included. Psychological distress encompassed depression, anxiety and anxiety-related disorders (e.g., social anxiety, panic disorder, PTSD, OCD), perceived stress, internalizing symptoms, suicidality and BPD symptoms, and negative affect. To align with our definition of distress, we included reports from clinical samples only if they were diagnosed with mental disorders classified under the fear and distress subfactors of the hierarchical taxonomy of psychopathology model (e.g., depressive disorders, anxiety disorders, PTSD, OCD, and BPD).

Effect sizes derived from studies using the TAS and DERS were inverted such that the negative correlation coefficient indicates that higher attention to emotion/emotional clarity is associated with lower psychological distress. In partial correlation analysis, studies were included when they reported all correlations between attention to emotion and emotional clarity *AND* between attention to emotion and distress *AND* between emotional clarity and distress, since extracting partial correlation from multiple regression including various predictors can introduce statistical bias (Aloe & Thompson, 2013).

Coding Procedure

The following variables were coded independently by the first and the second authors from each included study: (a) effect size(s), (b) sample size, (c) the type of measurements (for attention to emotion, 0 = DERS, 1 = TAS, 2 = TMMS, 3 = MAS, 4 = others; for emotional clarity, 0 = DERS, 1 = TAS, 2 = TMMS/Emotional Clarity Questionnaire, 3 = MAS, 4 = others), (d) sample characteristics (0 = community/college sample, 1 = psychiatric clinical sample, 2 = samples with physical conditions [e.g., breast cancer, multiple sclerosis], 3 = mixed sample of healthy and clinical participants), (e) gender (female ratio), (f) mean age, (g) study design (0 = cross-sectional, 1 = longitudinal), and (h) domains of psychological distress (0 = depression, 1 = anxiety, 2 = stress, 3 = negative affect, 4 = anxiety-related symptoms, 5 = suicidality and BPD symptoms, 6 = combined).

For the type of measurements, variants of the included scales and the WEFG were categorized as "others" due to their limited interpretability (e.g., a composite of the TMMS and the TAS) or the small number of effect sizes (e.g., $k = 3$ for the WEFG). For distress domains, symptoms of overall anxiety or specific anxiety disorders (e.g., generalized anxiety, panic, and social anxiety

³ Although the Attention to Emotion Scale also has an involuntary attention subscale, we chose to exclude this subscale from our analysis for consistency, as other measures primarily focus on voluntary aspects of attention to emotion. In addition, we excluded the Source of Emotions Scale (Boden & Berenbaum, 2012), which specifically examines one's understanding of the source of emotions, as other clarity measures assess one's understanding of the types of emotions experienced.

Table 1
Sample Items of Each Measurement

Scale	Item
TMMS	
Attention	I do not pay <i>much</i> attention to my feelings (r)
Clarity	I am usually very clear about my feelings
MAS	
Mood monitoring	I <i>often</i> evaluate my mood.
Mood labeling	I am never really sure what I am feeling (r)
TAS	
Externally oriented thinking	I find examination of my feelings useful in solving personal problems (r)
Difficulty identifying feelings	When I am upset, I do not know if I am sad, frightened, or angry
Difficulty describing feelings	I am able to describe my feelings easily (r)
DERS	
Lack of emotional awareness	I am attentive to my feelings (r)
Lack of emotional clarity	I have difficulty making sense out of my feelings
WEFG	
Attention to feelings	I pay attention to my feelings
Clarity of feelings	I have a hard time describing my feelings (r)

Note. The item with the highest factor loading for each subscale was selected from the references in the main text. TMMS = Trait MetaMood Scale; MAS = Mood Awareness Scale; TAS = Toronto Alexithymia Scale; DERS = Difficulties in Emotion Regulation Scale; WEF = Skalen zur Wahrnehmung eigener und fremder Gefühle; (r) = reverse-scored item.

disorders) were categorized under anxiety, whereas OCD and PTSD were classified as anxiety-related symptoms. When multiple domains were measured together (e.g., depression, anxiety, and stress computed as a total score), they were classified under the combined category. The κ coefficients for the recorded categorical variables ranged from 0.99 to 1.00. For the continuous variables, intraclass correlations ranged from 0.95 to 1.00. These results indicate excellent interrater agreement (Landis & Koch, 1977).

Statistical Analysis

All analyses were conducted in R Statistical Software (R Core Team, 2021) using “metafor” package (Viechtbauer, 2015). We used Pearson’s correlation coefficients (r) as our main effect size, which was transformed into Fischer’s Z in the analyses and reverted to r after the analysis for easier interpretation (Borenstein, 2009). Since majority of studies reported multiple effect sizes using different measures of emotional awareness/psychological distress (multiple dependent effect sizes were nested within samples), a multilevel model was suitable to address the dependency issue and to aggregate effect sizes (Cheung, 2014). The I^2 statistics were utilized to assess heterogeneity at Levels 2 and 3, while χ^2 likelihood-ratio tests were used to test whether each additional level of our model accounted for a statistically significant degree of variance in effect sizes. We performed analyses with separate models for attention to emotion and emotional clarity. Only Pearson’s zero-order correlations were collected for our study, as pooling partial and zero-order correlations together or transforming other coefficients into r can introduce statistical bias (Aloe & Thompson, 2013; Rupinski & Dunlap, 1996). A sensitivity analysis was conducted testing whether any outliers may change the weighted correlation based on Cook’s distances (Cook & Weisberg, 1982). A Cook’s distance > 4 times the mean suggested a value with high influence (Viechtbauer & Cheung, 2010).

For partial correlation analyses, partial correlations were computed to assess the relationship between attention to emotion and distress ($r_{p \text{ attention}}$) after controlling for clarity and the relationship

between emotional clarity and distress ($r_{p \text{ clarity}}$) after controlling for attention, with mathematical expression as follows:⁴

$$r_{p \text{ attention}} = \frac{r_{\text{attention}} - r_{\text{clarity}}r_{\text{corr}}}{\sqrt{(1 - r_{\text{clarity}}^2)}\sqrt{(1 - r_{\text{corr}}^2)}}, \quad (1)$$

$$r_{p \text{ clarity}} = \frac{r_{\text{clarity}} - r_{\text{attention}}r_{\text{corr}}}{\sqrt{(1 - r_{\text{attention}}^2)}\sqrt{(1 - r_{\text{corr}}^2)}}. \quad (2)$$

Next, we tested moderators, including the type of measurements, sample characteristics, age, gender, study design, and distress domains. An approximate R -squared (R^2 ; Harer et al., 2021) value was calculated for all significant moderators to determine the proportion of between-study variance explained by each moderator. Subgroup analyses were performed only if each category included at least six cases, as parameters can be poorly estimated with a small number of studies in a subgroup (e.g., Borenstein et al., 2021). Since nearly all studies with both psychiatric and physical clinical samples used the TAS or the DERS (51 out of 56), moderation analysis of sample characteristics was conducted using data from the subset of studies that used TAS and DERS. Last, to examine potential publication bias, visual inspection of funnel plots and Egger’s regression test were used. An extended version of Egger’s test for multilevel meta-analysis by including standard error as a moderator was applied (Fernández-Castilla et al., 2021).

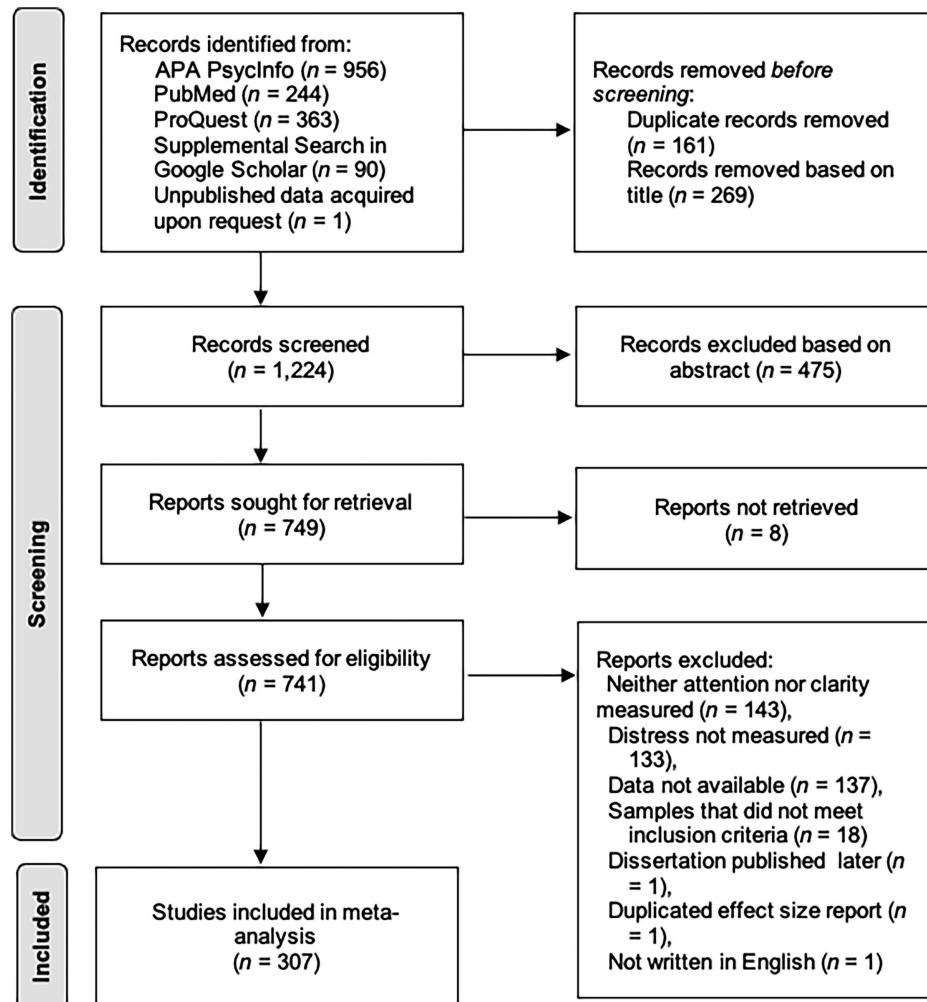
Transparency and Openness

The present study was conducted in compliance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 guidelines (Page et al., 2021). The protocol for the present study, including search criteria, inclusion criteria, and statistical methods for

⁴ In the formula, $r_{\text{attention}}$ is the correlation between attention to emotion and distress, r_{clarity} is the correlation between emotional clarity and distress, and r_{corr} is the correlation between attention to emotion and emotional clarity.

Figure 1

Preferred Reporting Items for Systematic Reviews and Meta-Analyses Flow Diagram for Literature Search



the main analysis, was preregistered at the Open Science Framework (Kim et al., 2025), where raw data and analysis code are also available. Moderation analyses and exclusion criteria were not part of the preregistration.

Results

Study Characteristics

The final data set consisted of 307 studies, including 615 effect sizes and a total of 100,612 participants. In cases where a single study provided effect sizes for multiple separate samples, each sample was treated as an individual study. Sample sizes ranged from 17 to 7,978. The average age of study participants was 29.94 years, with an average age range of 7.97–73.69 years. On average, 61.17% of participants were female, with percentages ranging from 0% to 100%. Out of 307 studies, 190 reported effect sizes for all three relationships—attention-distress, clarity-distress, and attention-clarity—and were included in the partial correlation analyses. Detailed characteristics of the included

articles, along with summary descriptions for these studies, are presented in the [Supplemental Materials](#).

Attention to Emotion and Psychological Distress

In the analysis, 485 effect sizes from 251 studies were included. The pooled correlation coefficient between attention to emotion and psychological distress was not significant, $r = -.01$, $p = .56$, 95% confidence interval (CI) $[-.03, .02]$. Thirty-seven outliers were identified. The sensitivity analysis, after excluding the outliers, produced a comparable result, $r = -.004$, 95% CI $[-.03, .02]$, $p = .76$. Significant heterogeneity was observed in the effect sizes, $Q(484) = 4,405.12$, $p < .001$, $I^2 = 90.33\%$. The majority of the variance in the effect sizes was attributed to between-study differences, $I^2 = 77.20\%$, with $I^2 = 13.13\%$ of within-study variance and 9.67% of total variance attributed to sampling error. The three-level model provided a significantly better fit compared to a two-level model with Level 3 heterogeneity constrained to 0, $\chi^2(1) = 216.41$, $p < .001$.

To investigate the relationship between attention to emotion and psychological distress while controlling for emotional clarity at the sample level, a meta-analysis was conducted again using partial correlations obtained from 190 included studies. The analysis revealed a small positive association between attention to emotion and psychological distress, $r_p = .11$, 95% CI [.09, .13], $p < .001$, $k = 360$, $I^2_W = 85.83\%$.

Moderators

The detailed results of significant categorical moderators, including pooled effect sizes for each category, are presented in Table 2. To facilitate interpretation, we reported—but did not compare—the effect size for the “others” category in the analysis of the type of measurements, as well as the effect size for the “combined” category in the analysis of the distress domain.

Type of Measurements. The moderation of the measurement type was significant, $F(4, 480) = 24.58$, $p < .001$. Specifically, in studies utilizing the TMMS/MAS, attention to emotion exhibited a small positive correlation with psychological distress, while the pooled effect size for the MAS was not statistically significant. Conversely, when measured using the DERS or TAS, attention to emotion was negatively correlated with psychological distress. Based on 95% CI, all pairwise comparisons between the TMMS/MAS (exhibiting a positive correlation) and the TAS/DERS (exhibiting a negative correlation)

were significant. Measurement type accounted for 36% of total between-study heterogeneity.

Sample Characteristics. Sample characteristics significantly moderated the relationship between attention to emotion and psychological distress, $F(2, 254) = 9.55$, $p < .001$. In the general community/college samples, the correlation was relatively weak. Conversely, studies involving psychiatric clinical samples and samples with physical conditions demonstrated a stronger negative association (see Table 2). Examination of the 95% CI showed that the relationship is stronger in psychiatric clinical samples compared to community/college samples. Sample characteristics accounted for 16% of the between-study heterogeneity in effect sizes.

Distress Domain. The measurements used in each distress domain category varied substantially (e.g., 38 out of 46 effect sizes measuring anxiety-related symptoms were assessed with the TAS/DERS). In addition, because both the direction and magnitude of the attention–distress relationship significantly varied by measurement type, we conducted two separate analyses: one for studies using the TMMS/MAS and another for those using the TAS/DERS. The distress domain was a significant moderator for both studies using the TMMS/MAS, $F(4, 184) = 3.66$, $p = .007$, and those using the TAS/DERS, $F(6, 254) = 3.54$, $p = .002$. For studies using the TMMS/MAS, the relationship between attention to emotion and distress was consistently small and positive across domains. For those using the TAS/DERS, the relationship was small and negative across domains. This

Table 2
Attention to Emotion: Significant Categorical Moderators

Moderator	<i>k</i>	<i>r</i> _{corrected} effect size	95% CI	<i>I</i> ² _W
Type of measurements				
TMMS	189	.10***	[.07, .13]	88.45%
MAS	6	.23 [†]	[−.03, .46]	83.57%
TAS	174	−.09***	[−.12, −.06]	81.97%
DERS	87	−.10***	[−.15, −.06]	86.38%
Others	29	.05	[−.04, .15]	87.96%
Sample characteristics				
TAS/DERS				
Community/college	183	−.06***	[−.09, −.04]	83.14%
Psychiatric	46	−.20***	[−.27, −.12]	75.11%
Physical conditions	28	−.12***	[−.16, −.08]	15.65%
Distress domains				
TMMS/MAS				
Depression	62	.09***	[.04, .15]	92.35%
Anxiety	61	.12***	[.07, .17]	88.63%
Stress	13	.07 [†]	[−.01, .15]	86.96%
Negative affect	45	.13***	[.07, .19]	86.13%
Anxiety-related	8	.09	[−.04, .21]	58.23%
TAS/DERS				
Depression	101	−.09***	[−.13, −.06]	82.57%
Anxiety	62	−.06**	[−.10, −.02]	73.08%
Stress	12	−.09*	[−.16, −.02]	62.75%
Negative affect	30	−.02	[−.08, .03]	73.06%
Anxiety-related	38	−.16***	[−.25, −.07]	92.38%
BPD/suicidality	10	−.12**	[−.19, −.05]	68.79%
Combined	8	−.20***	[−.26, −.14]	0.02%

Note. The following categories were excluded from the moderation analyses due to an insufficient number of studies: the “mixed” category ($k = 4$) in the sample characteristics analysis, the “BPD/suicidality” ($k = 4$) and “combined” ($k = 2$) categories in the distress domains analyses using the TMMS/MAS. CI = confidence interval; TMMS = Trait MetaMood Scale; MAS = Mood Awareness Scale; TAS = Toronto Alexithymia Scale; DERS = Difficulties in Emotion Regulation Scale; BPD = borderline personality disorder.

* $p < .05$. ** $p < .01$. *** $p < .001$. [†] $p < .10$.

moderator explained 0% of between-study heterogeneity and 27% of within-study heterogeneity in effect sizes measured with the TMMS/MAS, whereas it explained 3% of between-study heterogeneity in effect sizes measured with the TAS/DERS. Although moderation was significant, overlapping 95% CIs between domains in both analyses indicated no significant differences in effect sizes (see Table 2).

Study Design. Study design was not a significant moderator, $F(1, 483) = 2.21, p = .14$.

Age and Gender. Neither age nor gender was a significant moderator, $F(1, 433) = 1.30, p = .25$ for age, and $F(1, 456) = 0.04, p = .85$ for gender.

Emotional Clarity and Psychological Distress

For this model, 600 effect sizes from 298 studies were included. The meta-analysis of the relationship between clarity and distress yielded a moderate negative correlation, $r = -.33$, 95% CI $[-.35, -.32]$, $p < .001$. After removing 36 outliers, the results remained nearly consistent, $r = -.34$, 95% CI $[-.35, -.32]$, $p < .001$. Significant heterogeneity was observed in the effect sizes, $Q(599) = 5,731.19, p < .001$, $I^2 = 89.12\%$. Variance in effect sizes was primarily attributed to between-study level, $I^2 = 73.31\%$, while within-study level accounted for $I^2 = 15.81\%$, and 10.88% of total variance attributed to sampling error. The fit of the three-level model was significantly better than that of the two-level model, $\chi^2(1) = 255.28, p < .001$.

To examine the association between clarity and psychological distress while controlling for attention to emotion, we conducted another meta-analysis using partial correlations from 190 studies. The result showed a medium negative correlation after controlling for attention to emotion, $r_p = -.32$, 95% CI $[-.34, -.30]$, $p < .001$, $k = 360$, $I^2_W = 84.82\%$.

Moderators

The detailed results of the significant categorical moderators are presented in Table 3. Again, we reported the effect size for the “others” category in the analysis of measurement types but did not make comparisons.

Type of Measurements. Measurement type was a significant moderator, $F(4, 595) = 9.06, p < .001$, explaining 16% of the total between-study heterogeneity. While the negative relationships between emotional clarity and distress were significant across measurement types, the effect size was stronger in studies where clarity was measured using the TAS/DERS, compared to those using the TMMS/MAS, as indicated by the differences in the 95% CIs between these categories (see Table 3).

Sample Characteristics. Sample characteristics were not a significant moderator, $F(3, 323) = 1.59, p = .19$.

Distress Domain. For the same reasons as in the attention to emotion analysis, we conducted two separate analyses: one for articles using the TMMS/MAS and another for those using the TAS/DERS. Distress domains did not significantly moderate the relationship between clarity and distress for studies using the TAS/DERS, $F(6, 118) = 1.52, p = .18$. However, it was a significant moderator for studies using the TMMS/MAS, $F(4, 230) = 4.80, p = .001$, explaining 3% of the between-study heterogeneity in effect sizes. Despite significant moderation, small-to-medium negative correlations between clarity and distress were found across

Table 3

Emotional Clarity: Significant Categorical Moderators

Moderator	<i>k</i>	<i>r</i> _{corrected} effect size	95% CI	<i>I</i> ² _W
Type of measurements				
TMMS	235	-.28***	$[-.30, -.25]$	86.74%
MAS	6	-.18*	$[-.31, -.03]$	58.31%
TAS	202	-.38***	$[-.40, -.35]$	83.70%
DERS	125	-.39***	$[-.43, -.35]$	91.54%
Others	32	-.29***	$[-.37, -.21]$	87.89%
Distress domains				
TMMS/MAS				
Depression	96	-.31***	$[-.34, -.28]$	84.14%
Anxiety	69	-.27***	$[-.31, -.23]$	87.37%
Stress	13	-.33***	$[-.39, -.27]$	80.99%
Negative affect	47	-.23***	$[-.29, -.17]$	87.34%
Anxiety-related	10	-.18**	$[-.28, -.08]$	62.80%
Study designs				
Cross-sectional	557	-.34***	$[-.35, -.32]$	88.86%
Longitudinal	43	-.25***	$[-.33, -.17]$	90.59%

Note. The “BPD/suicidality” ($k = 4$) and “combined” ($k = 2$) categories in the distress domain analysis were excluded due to an insufficient number of studies. CI = confidence interval; TMMS = Trait MetaMood Scale; MAS = Mood Awareness Scale; TAS = Toronto Alexithymia Scale; DERS = Difficulties in Emotion Regulation Scale; BPD = borderline personality disorder.

* $p < .05$. ** $p < .01$. *** $p < .001$.

distress domains, with overlapping 95% CIs, indicating nonsignificant differences between categories. Detailed results are presented in Table 3.

Study Design. The study design significantly moderated the relationship between clarity and distress, $F(1, 598) = 10.46, p = .001$, explaining 1% of the total between-study heterogeneity. Specifically, the relationship between emotional clarity and distress was stronger in cross-sectional studies compared to longitudinal ones. Again, however, the 95% CIs overlapped between the two categories.

Age and Gender. Neither age nor gender emerged as a significant moderator, $F(1, 542) = 0.004, p = .95$ for age, and $F(1, 558) = 0.75, p = .39$ for gender.

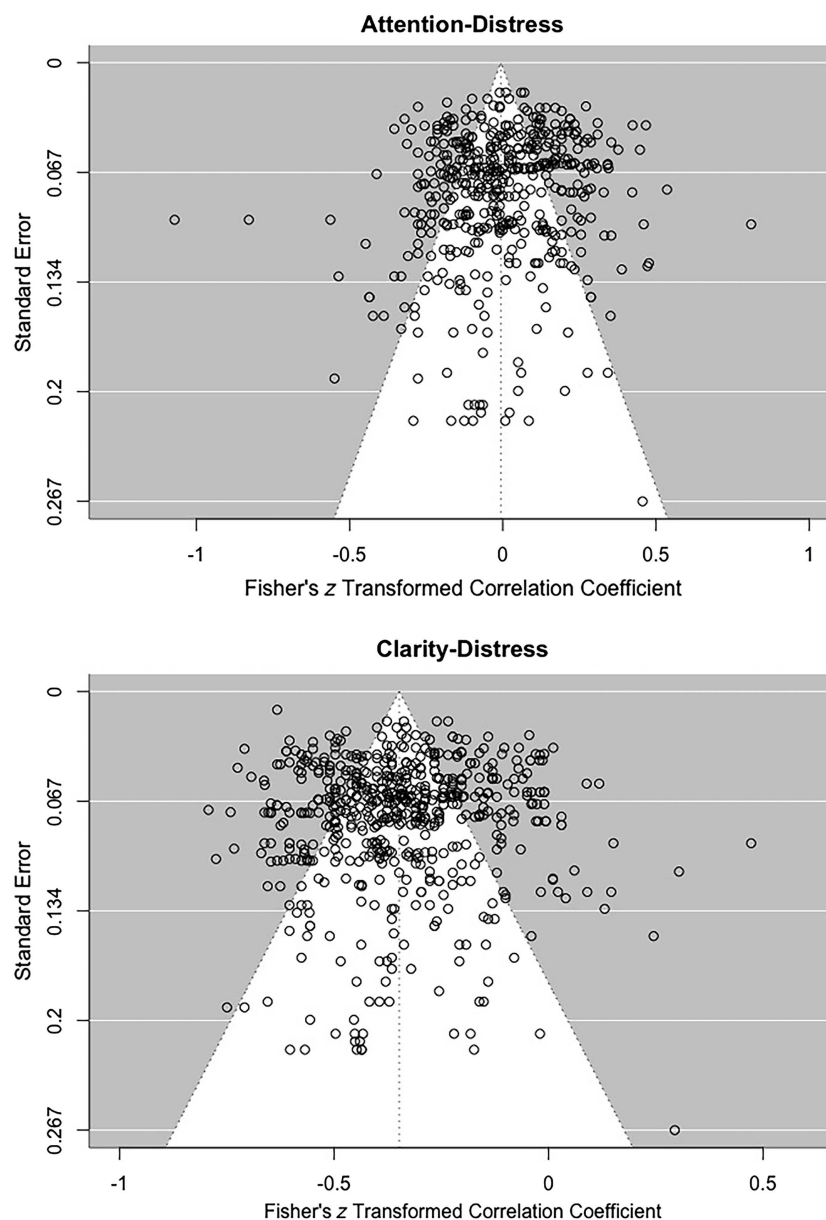
Publication Bias

Potential publication bias was examined by visual inspection of funnel plots and Egger’s regression test. These two techniques assume that if publication bias operates, small studies (i.e., studies with smaller sample sizes) tend to report larger effect sizes, creating asymmetrical distributions of studies when plotting them with effect sizes on the X-axis and variance on the Y-axis. Upon visual inspection, the funnel plots seemed relatively symmetrical (see Figure 2). Egger’s regression test indicated that the asymmetry in the funnel plots was not statistically significant, $t(483) = -1.62, p = .11$, for the attention-distress analysis, $t(598) = 1.05, p = .29$, for the clarity-distress analysis.

Discussion

This meta-analysis aimed to clarify the distinct relationship between two facets of emotional awareness—attention to emotion and emotional clarity—and psychological distress. We first examined the overall relationships between each facet and psychological

Figure 2
Funnel Plots for Effect Sizes in the Meta-Analyses



distress, then investigated whether these facets were uniquely associated with distress using partial correlation analysis. Our main findings were as follows: attention to emotion was not significantly correlated to psychological distress, while emotional clarity had a moderate negative correlation with psychological distress. The relationship between emotional clarity and distress remained consistent when controlling for attention to emotion. However, when accounting for emotional clarity, the relationship between attention and psychological distress became significantly positive. In addition, the relationships between distress and the two facets of emotional awareness, especially attention to emotion, were significantly moderated by the type of measurement.

Overall, the distinct relationships between the two facets of emotional awareness and psychological distress found in this study highlight the importance of examining and understanding attention to emotion and emotional clarity as separate domains rather than as a composite construct of emotional awareness when studying their relation to psychological distress. For example, a widely used measure of alexithymia, the TAS, consists of one subscale measuring attention to emotion (externally oriented thinking) and two subscales measuring emotional clarity (difficulties identifying/describing feelings). However, it has often been treated as a unified construct by using a composite score when studying its relation to distress (e.g., suicidality; see Hemming et al., 2019). Our findings advocate for

examining attention to emotion and emotional clarity as distinct factors, especially when they are components of broader constructs such as alexithymia, emotional awareness, metamood, or emotion dysregulation, to better understand their unique contributions to psychological distress.

More specifically, our finding of a moderate negative relationship between emotional clarity and distress is consistent with previous meta-analyses (S. Li et al., 2015; Visted et al., 2018) as well as mood and emotion regulation theories, which posit that understanding one's emotions is essential for effective regulatory processes (Gross, 2015; Larsen, 2000). Emotional clarity may result in decreased distress via more effective/successful emotion regulation. For example, previous studies revealed that greater emotional clarity was associated with lower levels of rumination about the causes and state of one's feelings (e.g., Rubenstein et al., 2015; Vine et al., 2014). It is possible that a clear understanding of one's emotions may help conserve cognitive resources, allowing them to be invested in selecting and implementing regulatory strategies that fit well with a given context (Hemenover et al., 2008). In addition, individuals with higher emotional clarity may feel more confident in managing their emotions, leading to greater regulatory efforts and, consequently, a higher likelihood of successful mood and emotion regulation. Indeed, those with higher levels of clarity tended to report greater self-efficacy in emotion regulation (Kämpfe & Mitte, 2010), and greater emotion regulation self-efficacy was linked to less negative emotional responses to negative stimuli (Benfer et al., 2018).

In contrast, the relationship between attention to emotion and psychological distress was not significant. However, these results drastically shifted when accounting for emotional clarity, as well as the inclusion of moderators, especially measurement type. The relationship between attention to emotion and distress became significantly positive when controlling for emotional clarity, while the clarity–distress relationship remained comparable when accounting for attention to emotion. Individuals with high emotional clarity are likely to pay some degree of attention to their emotions, as attention is necessary for understanding them. However, despite paying significant attention to their emotions, some individuals may still struggle with emotional clarity.

These results, which show a positive association between attention to emotion and distress when controlling for emotional clarity, suggest that the benefits of attention to emotion in emotional health may be specifically linked to emotional clarity. Previous studies have indicated that attention to emotion may involve both adaptive and maladaptive/ruminative aspects of self-focus (e.g., Lischetzke & Eid, 2003), and emotional clarity may play a crucial role in the relationship between attention to internal states and distress (e.g., K. A. Coffey et al., 2010; Salguero et al., 2013). For example, K. A. Coffey et al. (2010) found that attention to internal states (e.g., sensations; thoughts, emotions) had a negative indirect relationship with psychological distress through emotional clarity, while the direct correlation between attention and distress was positive. Further, Salguero et al. (2013) demonstrated that attention to emotion is linked to depression through two pathways: one pathway involves rumination, which was associated with increased depression, while the other pathway involves emotional clarity, which was related to decreased depression. Along with these findings, our results suggest that the potential benefits of attention to emotion in reducing distress may be linked to its role in enhancing emotional clarity. Attention

to emotion itself, when controlling for this potential benefit, was associated with increased distress, potentially via rumination.

In addition, the type of measurement significantly moderated the relationship between distress and the two facets, particularly attention to emotion, as its direction varied depending on the measurement used. Specifically, attention to emotion was negatively associated with distress when measured by the TAS and the DERS but positively associated with distress when measured by the TMMS. Each scale appears to focus on different aspects of attention to emotion. The externally oriented thinking subscale of the TAS (e.g., "Being in touch with emotions is essential") and the lack of awareness subscale of the DERS (e.g., "I care about what I am feeling") primarily address the valuation of feelings. In contrast, the attention subscale of the TMMS (e.g., "I pay a lot of attention to my feelings") and the monitoring subscale of the MAS (e.g., "I often evaluate my mood") focus more on the extent to which a person attends to their emotions. Extremely high scores on the TMMS or the MAS may indicate an excessive focus on inner experiences, whereas moderate levels might reflect an adequate awareness.

Paying excessive attention to emotion can hinder flexible interchange between domains of self-regulation (Tamir, 2021). When an individual becomes overly focused on their internal emotional state, it might impair their ability to shift attention to other critical aspects of their environment. This excessive focus on emotions can disrupt the overall functioning of the individual, as it hampers the necessary balance between various regulatory domains essential for adaptiveness. Our results suggest that valuing and acknowledging emotions as an important source of information, as measured by the DERS and/or the TAS, might be the beneficial aspect of paying attention to emotion in managing distress, particularly in clinical populations, while excessive attention might lead to maladaptive self-focus.

Relatedly, the positive relationship between distress and attention to emotion, as measured by the TMMS and the MAS, might be better represented by a curvilinear relationship, as moderate levels of attention to internal states could be theoretically beneficial. De la Barrera et al. (2023) found both significant linear and curvilinear relationships between attention to emotion, as measured by the TMMS, and perceived stress. Further research is needed to clarify whether these relationships are linear or curvilinear, particularly for attention to emotion as measured by the TMMS and the MAS, and their impact on psychological distress.

We believe that the importance of our study lies in its clinical implications. Promoting a person's awareness of their emotions (both paying attention to and identifying emotions) has been a central focus in various psychological treatments, such as dialectical behavior therapy (Linehan et al., 1999), emotion-focused therapy (Greenberg, 2004), and mindfulness-based cognitive behavioral therapy (Segal et al., 2012). However, the different implications of attention to emotion and emotional clarity, as well as the potential pitfalls of excessively focusing on emotions (as noted by Davis & Nichols, 2016), have remained understudied. Our results suggest that paying attention to emotion may not always be beneficial. Its effectiveness may depend on whether it is accompanied by a clear understanding of emotions and the appropriate degree of attention. It could be crucial to help clients clearly understand and identify their emotions while paying appropriate attention to them, without becoming excessively focused on their internal states. Monitoring how much attention a person pays to their internal states could help prevent it from turning into counterproductive rumination without achieving clarity. Indeed,

Butler et al. (2018) demonstrated that only emotional clarity, not attention to emotion, was a significant mediator of therapy outcomes.

Our study is not without its limitations. First, due to the correlational nature of our data, causal relationships between each facet of emotional awareness and distress cannot be confirmed. While we also examined longitudinal relationships—finding a significant association between clarity and distress—a longitudinal design alone is insufficient to establish causality, highlighting the need for future experimental research. Although previous intervention studies have not specifically targeted each facet of emotional awareness, interventions aimed at enhancing general emotional awareness have been shown to reduce negative affect, particularly among individuals with low attention to emotion (Baker & Berenbaum, 2008, 2011). Further, Sánchez-Álvarez et al. (2019) conducted a cross-lagged panel analysis with annual assessments over 3 years to examine the temporal directionality between emotional awareness and distress. They found that both emotional clarity and attention to emotion more consistently predicted future changes in negative affect, rather than the reverse, although attention to emotion predicted increases in negative affect in their sample. These results suggest the possibility of a causal relationship between the two facets of emotional awareness and distress.

Second, the present study did not review all types of emotional awareness identified in the literature. Previous research has distinguished between different dimensions of each facet, such as source versus type clarity (understanding the source or cause of one's emotions vs. the types of emotions experienced) and voluntary versus involuntary attention to emotion (controlled vs. automatic attention; see Boden & Berenbaum, 2012, for emotional clarity facets; Huang et al., 2013, for attention to emotion facets). This review did not include involuntary attention and source clarity, as the relevant questionnaires were not part of this analysis due to insufficient research, particularly in relation to distress. Prior findings suggest that examining these additional facets is crucial for understanding distress. For example, source clarity has been associated with fewer general psychiatric symptoms (Millgram et al., 2023) and less frequent suicidal thoughts (Millgram et al., 2025). In addition, involuntary attention to emotion—but not voluntary attention—has been linked to greater depression (e.g., Boden & Thompson, 2015; Eckland et al., 2021). This indicates that future research should aim to measure and investigate these distinct facets in relation to distress.

Last, unexplained heterogeneity was present across all analyses, despite testing several potential moderators. One potential explanation is the interplay between facets of emotional awareness, where each facet's relationship with distress depends on the level of the other. Our partial correlation analysis revealed that controlling for emotional clarity altered the link between attention to emotion and distress, suggesting that clarity may influence this relationship. Supporting this, previous studies have shown that individuals with similar levels of attention experience report different distress levels based on their emotional clarity (Berenbaum et al., 2012; Sperry & Eckland, 2021). Similarly, interventions mainly targeting emotional clarity influenced distress differently depending on baseline attention to emotion levels (Baker & Berenbaum, 2008, 2011). Overall, these findings highlight the need for future research to directly test whether the relationship between each facet of emotional awareness and distress is moderated by the other facet.

Nevertheless, this study represents the first quantitative synthesis of the distinct relationship between the two facets of emotional awareness and psychological distress, encompassing a broad range of relevant measures and constructs, with a substantial number of individual studies included. Our findings expand the literature by demonstrating that emotional clarity has a stronger association with psychological distress, whereas the relationship between attention to emotion and distress is weaker and varies significantly depending on a covariate (emotional clarity) and moderators. These results highlight the potential importance of addressing emotional clarity when helping individuals enhance their emotional awareness and reduce psychological distress.

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